

**SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ**

COURSE NAME: COMPUTER VISION

COURSE CODE:ITA0515

Slot B

COURSE OUTCOME COVERED

CO3: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)

Assessment Tool Weightage: 10%

QUESTIONS:50

Total Marks 50

Duration :60 Minutes

Annexure A

MCQ

Code of Conduct:

I K. Lahari (192325118) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Lahari

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect	
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers	
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts	
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time	
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions	

1. Image Normalization (Preprocessing)

An image pixel value $r = 150$ is normalized to the range $[0,1]$ from **8-bit** data.

What is the normalized value?

- A) 0.39
- B) 0.49
- C) 0.59
- D) 0.78

Ans: C-0.59

2. Image Storage Calculation

A grayscale image of size 1024×512 is stored using **8 bits per pixel**.

What is the storage requirement?

- A) 256 KB
- B) 512 KB
- C) 1024 KB
- D) 2048 KB

Ans: B-512KB

3. Edge Detection – Gradient Magnitude

Given Sobel gradients:

($G_x = 6$), ($G_y = 8$)

What is the gradient magnitude?

- A) 10
 - B) 12
 - C) 14
 - D) 16
- Ans:** A-10

4. Edge Direction Calculation

For the same gradients ($G_x = 6$), ($G_y = 8$),
what is the **edge direction**?

- A) 36.8°
 - B) 45°
 - C) 53.1°
- Ans:** C-53.1

5. Harris Corner Detection Response

Given:

($R = \det(M) - k(\text{trace}(M))^2$)

Where:

$\det(M) = 200$, $\text{trace}(M) = 30$, $k = 0.04$

What is the corner response R ?

A) 136

B) 164

C) 180

D) 200

Ans:B-164

6. Harris Corner Classification

If the Harris response $R = +250$, the pixel is classified as:

A) Flat region

B) Edge

C) Corner

D) Noise

Ans:C

7. Hough Transform – Line Representation

A line detected at $\theta = 90^\circ$ and $p = 50$ corresponds to:

A) Horizontal line at $y = 50$

B) Vertical line at $x = 50$

C) Diagonal line

D) Random line

Ans:A

An image is scaled by a factor of **2**.

If original keypoint scale is $\sigma = 1.6$, new scale is:

A) 0.8

B) 1.6

C) 3.2

D) 4.0

Ans:C

9. PCA Dimensionality Reduction

A feature vector has **100 dimensions**.

PCA retains **95% variance using 20 components**.

What is the dimensionality after PCA?

A) 5

B) 20

C) 80

D) 95

Ans:B

10. PCA Covariance Matrix Size

For a dataset with **50 features**, what is the size of the covariance matrix?

- A) 50×1
- B) 1×50
- C) 50×50
- D) 100×100

Ans:C

11. Image Quantization (Preprocessing)

A grayscale image is quantized from **256 levels to 32 levels**.

What is the **number of bits per pixel after quantization**?

- A) 3
- B) 4
- C) 5
- D) 6

Ans:C

12. Image Storage Reduction

An RGB image of size **512×512** uses **8 bits per channel**.

It is converted to grayscale with **8 bits per pixel**.

What is the **percentage reduction in storage**?

- A) 33%
- B) 50%
- C) 66.7%
- D) 75%

Ans:C

13. Sobel Edge Response

Given Sobel kernel response values:

($G_x = -12$), ($G_y = 5$)

What is the **gradient magnitude (approx.)**?

- A) 10
- B) 12
- C) 13
- D) 15

Ans:C

14. Gradient Direction Calculation

For ($G_x = -12$), ($G_y = 5$), the gradient direction is closest to:

- A) 22°
- B) 157°

- C) 180°
- D) -22°

Ans:B

15. Harris Corner Detection Response

Given:

$$\det(M) = 400$$

$$\text{trace}(M) = 40$$

$$k = 0.06$$

Compute the Harris response **R**.

- A) 256
- B) 304
- C) 352
- D) 400

Ans:B

16. Harris Corner Decision

For a pixel, eigenvalues of matrix **M** are:

$$\lambda_1 = 120, \lambda_2 = 130$$

This pixel is classified as:

- A) Flat region
- B) Edge
- C) Corner
- D) Noise

Ans:C

17. Hough Transform Voting

A point contributes votes to **180 values of θ** in Hough space.

What is the **shape of its representation**?

- A) A point
- B) A straight line
- C) A sinusoidal curve
- D) A circle

Ans:C

18. Hough Line Detection

A detected line has parameters **$p = 0, \theta = 45^\circ$** .

Which line is represented?

- A) Horizontal through origin
- B) Vertical through origin
- C) Diagonal line through origin

D) Random orientation

Ans:C

19. SIFT Keypoint Descriptor Size

Each SIFT descriptor consists of **4×4 histograms**, each with **8 bins**.

What is the descriptor dimensionality?

A) 64

B) 96

C) 128

D) 256

Ans:C

20. PCA Eigenvalue Selection

Eigenvalues of covariance matrix:

[40, 25, 15, 10, 5, 5]

How many principal components are needed to retain **80% variance**?

A) 2

B) 3

C) 4

D) 5

Ans:B

21. Preprocessing – Normalization Scenario

A surveillance camera outputs **8-bit grayscale images**.

Before feature extraction, pixel values must be normalized to **[0,1]**.

A pixel has value **204**.

What is the normalized value?

A) 0.60

B) 0.70

C) 0.80

D) 0.90

Ans:C

22. Image Representation Scenario

A medical image of size **2048 × 1024** is stored using **16 bits per pixel**.

What is the **memory required**?

A) 2 MB

B) 4 MB

C) 8 MB

D) 16 MB

Ans:B

23. Edge Detection Scenario

At a pixel location, Sobel gradients are:

($G_x = 9$), ($G_y = 12$)

What is the **edge magnitude** at that pixel?

- A) 15
- B) 18
- C) 21
- D) 25

Ans:A

24. Edge Orientation Scenario

Using the same gradients ($G_x = 9$), ($G_y = 12$),
what is the **edge direction (approx.)**?

- A) 36.8°
- B) 45°
- C) 53.1°
- D) 60°

Ans:C

25. Harris Corner Detection Scenario

For a region in an image, the Harris matrix values are:

$\det(M) = 300$, $\text{trace}(M) = 25$, $k = 0.04$

What is the **Harris response R**?

- A) 200
- B) 250
- C) 275
- D) 300

Ans:C

26. Harris Corner Interpretation Scenario

If the Harris response **R is large and positive**, the pixel corresponds to:

- A) Flat region
- B) Edge
- C) Corner
- D) Noise

Ans:C

27. Hough Transform Scenario

A point in image space votes for all possible **θ values** in Hough space.

What shape does this vote form?

- A) A point
- B) A straight line
- C) A sinusoidal curve
- D) A circle

Ans:C

28. Hough Line Detection Scenario

A detected line has parameters $\rho = 40$, $\theta = 90^\circ$.

Which line is represented?

- A) Vertical line $x = 40$
- B) Horizontal line $y = 40$
- C) Diagonal line

Ans:B

29. SIFT Feature Scenario

An image is scaled by a factor of **0.5**.

A SIFT keypoint detected at scale $\sigma = 2.0$ originally will now appear at:

- B) 0.5
- A) 1.0
- C) 2.0
- D) 4.0

Ans:A

30. PCA Dimensionality Reduction Scenario

A feature vector has **200 dimensions**.

After PCA, **30 principal components** retain **92% variance**.

What is the **new feature dimension**?

- A) 30
- B) 92
- C) 170
- D) 200

Ans:A

31. Preprocessing – Bit Depth Conversion

A grayscale image originally stored in **12-bit format** is converted to **8-bit** before processing.

An original pixel value is **3000**.

What is the approximate 8-bit value after linear scaling?

- A) 120
- B) 150
- C) 187
- D) 225

Ans:C

32. Image Representation – Sampling Scenario

A camera samples an image at **600 × 400** pixels.

The image is down-sampled by a factor of **2 in both directions**.

What is the new image size?

- A) 150×100
- B) 300×200
- C) 400×300
- D) 600×400

Ans:B

33. Edge Detection – Sobel Response

At a pixel, Sobel gradients are:

($G_x = -8$), ($G_y = 15$)

What is the **edge magnitude (approx.)**?

- A) 13
- B) 15
- C) 17
- D) 20

Ans:C

34. Edge Orientation Scenario

For ($G_x = -8$), ($G_y = 15$), the gradient direction is closest to:

- A) 28°
- B) 62°
- C) 118°
- D) 152°

Ans:C

35. Harris Corner Detection – Computation

Given:

$\det(M) = 500$, $\text{trace}(M) = 35$, $k = 0.05$

What is the Harris response **R**?

- A) 439
- B) 450
- C) 469

Ans:A

36. Harris Corner Classification

If one eigenvalue is large and the other is small, the pixel represents:

- A) Flat region
- B) Edge
- C) Corner
- D) Noise

Ans:B

37. Hough Transform – Line Interpretation

A line detected with parameters $p = 0$, $\theta = 90^\circ$ corresponds to:

- A) Vertical line at $x = 0$
- B) Horizontal line at $y = 0$
- C) Diagonal line

Ans:B

38. SIFT Descriptor Scenario

Each SIFT descriptor contains 4×4 blocks with **8 orientation bins** each.

How many values describe one keypoint?

- A) 32
- B) 64
- C) 96

Ans:C

39. PCA Variance Retention

Eigenvalues of covariance matrix are:

[50,; 30,; 10,; 5,; 5]

How many principal components are needed to retain **90% variance**?

- A) 1
- B) 2
- C) 3
- D) 4

Ans:C

40. PCA Projection Scenario

Original feature vector dimension = **120**.

PCA reduces it to **25 components**.

What is the **reduction percentage**?

- A) 60%
- B) 70%

C) 79%

Ans:C

41. Quantization Error Analysis (Preprocessing)

An 8-bit grayscale image is quantized to **64 levels**.
Original pixel value = **150**.

What is the **maximum possible quantization error**?

- A) ± 1
- B) ± 2
- C) ± 4
- D) ± 8

Ans:B

42. Image Representation Storage Comparison

Image A: **1024 × 1024 grayscale (8-bit)**

Image B: **1024 × 1024 RGB (8-bit per channel)**

What is the **storage ratio (B : A)**?

- A) 1 : 1
- B) 2 : 1
- C) 3 : 1
- D) 4 : 1

Ans:C

43. Sobel Edge Strength Comparison

At two pixels:

Pixel-1: ($G_x = 6$,; $G_y = 8$)

Pixel-2: ($G_x = 9$,; $G_y = 12$)

Which pixel corresponds to a **stronger edge**?

- A) Pixel-1
- B) Pixel-2
- C) Both equal

Ans:B

44. Harris Corner Response Comparison

Two regions have the following Harris parameters:

Region **det(M)** **trace(M)**

R1 300 25

Region det(M) trace(M)

R2 350 40

Given $k = 0.04$, which region is **more likely a corner**?

- A) R1
- B) R2
- C) Both
- D) Neither

Ans:B

45. Harris Corner vs Edge Decision

Eigenvalues of matrix M at a pixel are:

$\lambda_1 = 120, \lambda_2 = 10$

What does this pixel represent?

- A) Flat region
- B) Edge
- C) Corner
- D) Noise

Ans:B

46. Hough Transform Voting Analysis

Two points generate curves in Hough space that intersect at $(\rho = 60, \theta = 45^\circ)$.

What does this intersection indicate?

- A) A corner
- B) A circle
- C) A line in image space
- D) Noise accumulation

Ans:C

47. Hough Transform Parameter Effect

Increasing the θ resolution from 1° to 0.5° results in:

- A) Fewer accumulator cells
- B) Lower memory usage
- C) Higher line detection accuracy

Ans:C

48. SIFT Scale Analysis

An object appears at scale $\sigma = 1.6$ in Image-1.

In Image-2, the object is scaled by $1.5\times$.

At what scale should the corresponding SIFT keypoint appear?

- A) 1.1
- B) 1.6
- C) 2.4

Ans:C

49. PCA Variance Contribution Analysis

Eigenvalues of covariance matrix are:

[60,, 20,, 10,, 5,, 5]

How many principal components are needed to retain **90% variance**?

- A) 1
- B) 2
- C) 3
- D) 4

Ans:C

50. PCA Dimensionality Reduction Impact

Original feature dimension = **150**

After PCA = **30 components**

What is the **percentage reduction**?

- A) 60%
- B) 70%
- C) 80%
- D) 85%

Ans:C